



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Selected Topics in Cryptography

Lab 07: Message Authentication Code

April 9, 2025

1. Research about Message Authentication Codes

Look for information in the web page of NIST <https://csrc.nist.gov/projects/message-authentication-codes> and answer the following questions before implement any MAC in software. Write down a brief report by your own, with the answers.

1. According to NIST, which are the approved general purpose MAC algorithms?
2. Which is the standard for HMAC?
3. Using the standard for HMAC, describe which are the parameters required and how it works.
4. Find the security level for HMAC.
5. Which is the standard for KMAC?
6. Which is the standard for CMAC?
7. According to NIST standard for KMAC answer the following questions:
 - what is KECCAK Message Authentication Code?
 - which are the four fixed-length hash functions ?
 - what is an extendable-output function (XOF)?
 - which are eXtendable Output Functions ?
 - what is a sponge function?
8. Which are the cSHAKE variants describe in the NIST standard? What is the security strength for each of them?
9. According to NIST standard, how does CMAC work? Does it use any other cryptographic algorithm? Which one?
10. According to NIST standard how does it make MAC generation?

2. Programming exercises

Please choose a cryptographic library that implements a MAC specified in the NIST standard and follow the steps given in class: key generation, tag generation (also called MAC generation), and verification. Upload photos of your source code and screenshots of your program running.

3. Products

Every student must write her/his own report including the following information:

1. Your personal information, date of the lab session and the topic that we are studying in this lab session.
2. Answers for questions of section I